

COMMUNITY LIBRARY MANAGEMENT -DATABASE DESIGN

NAME: KEERTHIKESWARAN

1. Admins: Stores administrative users who manage the system.

AdminId: Integer (Primary Key)

AdminName: String

AdminPhone: String (Unique)

AdminEmail: String (Unique)

2. Members: Stores the library members who borrow books.

MemberId: Integer (Primary Key)

MemberName: String

MemberPhone: String

MemberEmail: String

MemberType: String (Basic, Premium, Student)

MemberStatus: String (Active, Inactive)

3. Books: Stores the metadata for book titles.

BookId: Integer (Primary Key)

BookTitle: String

BookAuthor: String

BookCategory: String

BookContents: String (Long Text)

CategoryNumber: Integer (Foreign Key to BookCategories)

4. BookCategories: Stores distinct categories to organize books.

CategoryNumber: Integer (Primary Key)

CategoryName: String

5. BookCopies: Tracks individual physical copies of a book title.

CopyId: Integer (Primary Key)

BookId: Integer (Foreign Key to Books)

CopyStatus: String (Available, Borrowed, Unavailable)

CopyCondition: String (Good, Damaged)

6. Borrowings: Logs instances when a member borrows a specific book.

BorrowId: Integer (Primary Key)

MemberId: Integer (Foreign Key to Members)

BookId: Integer (Foreign Key to Books)

BorrowDate: Date

DueDate: Date

ReturnStatus: String (Borrowed, Returned)

Remarks: String

7. Returns: Tracks the details of a book being brought back, pending admin approval.

ReturnId: Integer (Primary Key)

BorrowId: Integer (Foreign Key to Borrowings)

ActualReturnDate: Date

FineAmount: Float

ReturnApprovalStatus: String (Approved, Pending)

8. FineCalculation: Stores penalty records applied to a member.

FineConfigId: Integer (Primary Key)

BorrowId: Integer (Foreign Key to Borrowings)

FineAmount: Float

FineStatus: String (Paid, Unpaid)

Remarks: String (Late, Damaged, Both)

9. MembershipLimits: Defines rules based on the member tier.

MemberType: String (Primary Key)

MaxBooksAllowed: Integer

BorrowDurationDays: Integer

10. FineConfiguration: Stores the dynamic rates for different penalties.

FineId: Integer (Primary Key)

FineType: String (e.g., LateReturn)

Amount: Float

Relationships

Members (1) -----> (Many) Borrowings

Books (1) -----> (Many) Borrowings

BookCategories (1) -----> (Many) Books

Books (1) -----> (Many) BookCopies (Cascade)

Borrowings (1) -----> (Many) Returns

Borrowings (1) -----> (Many) FineCalculation

MembershipLimits (1) -----> (Many) Members